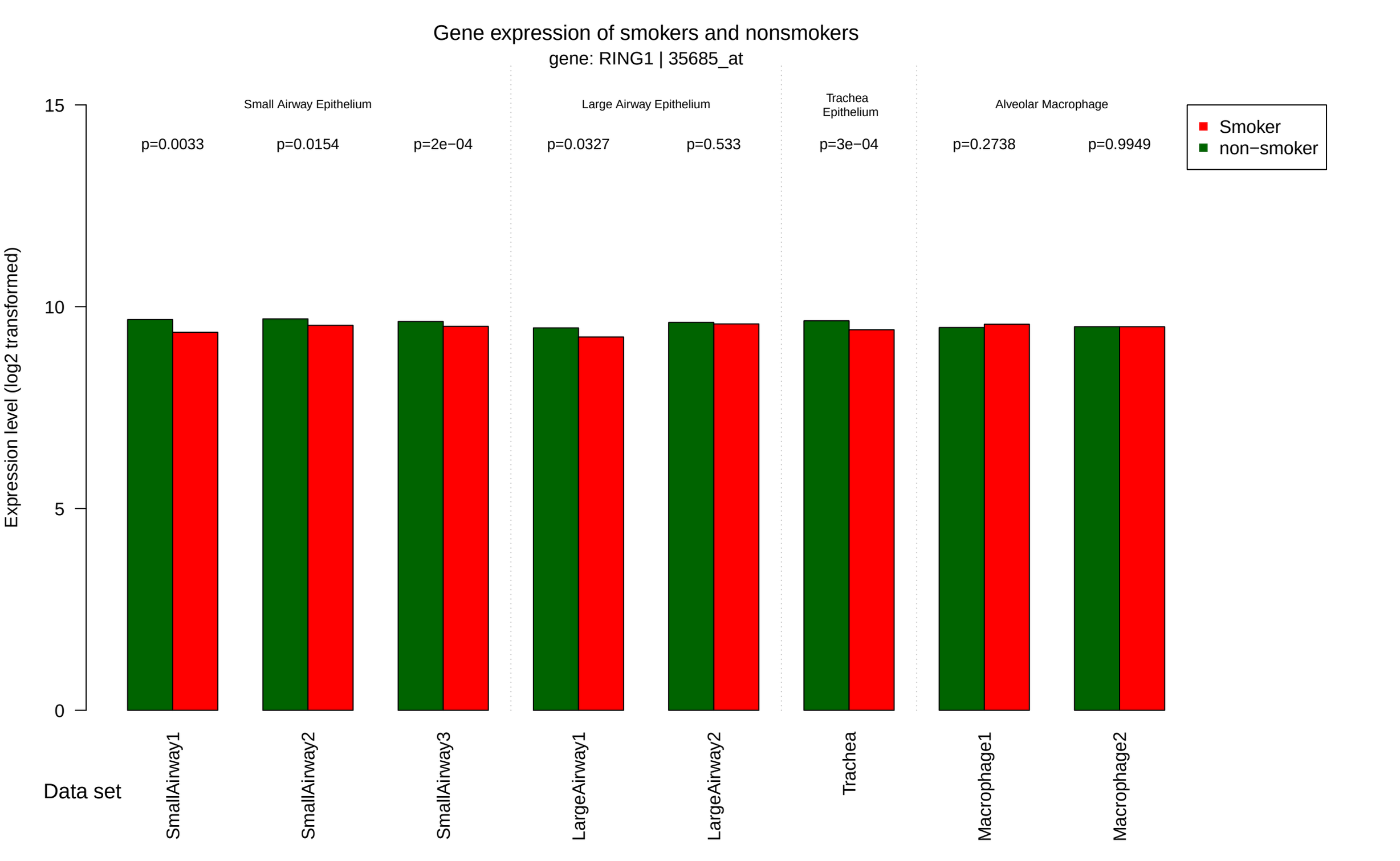
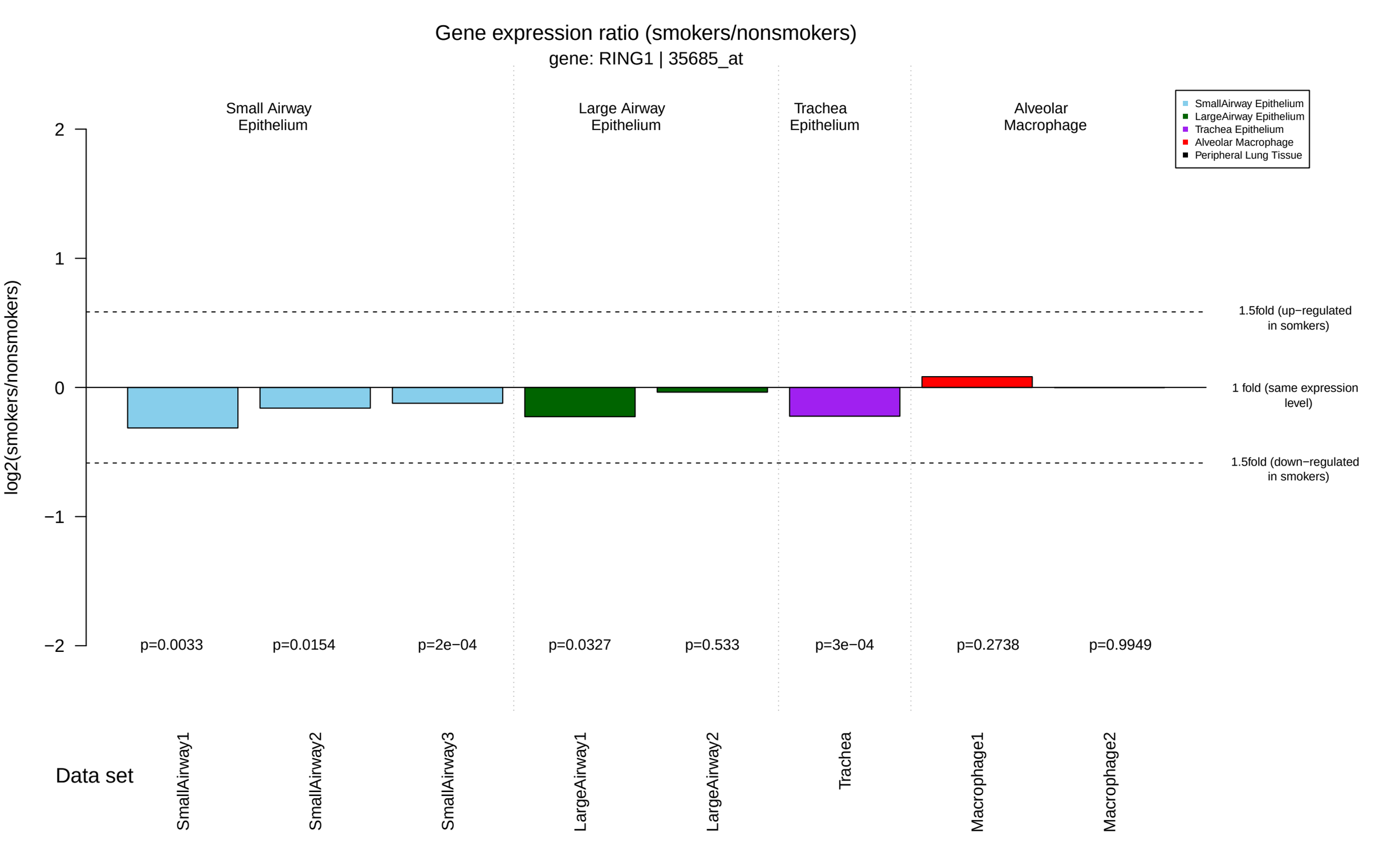
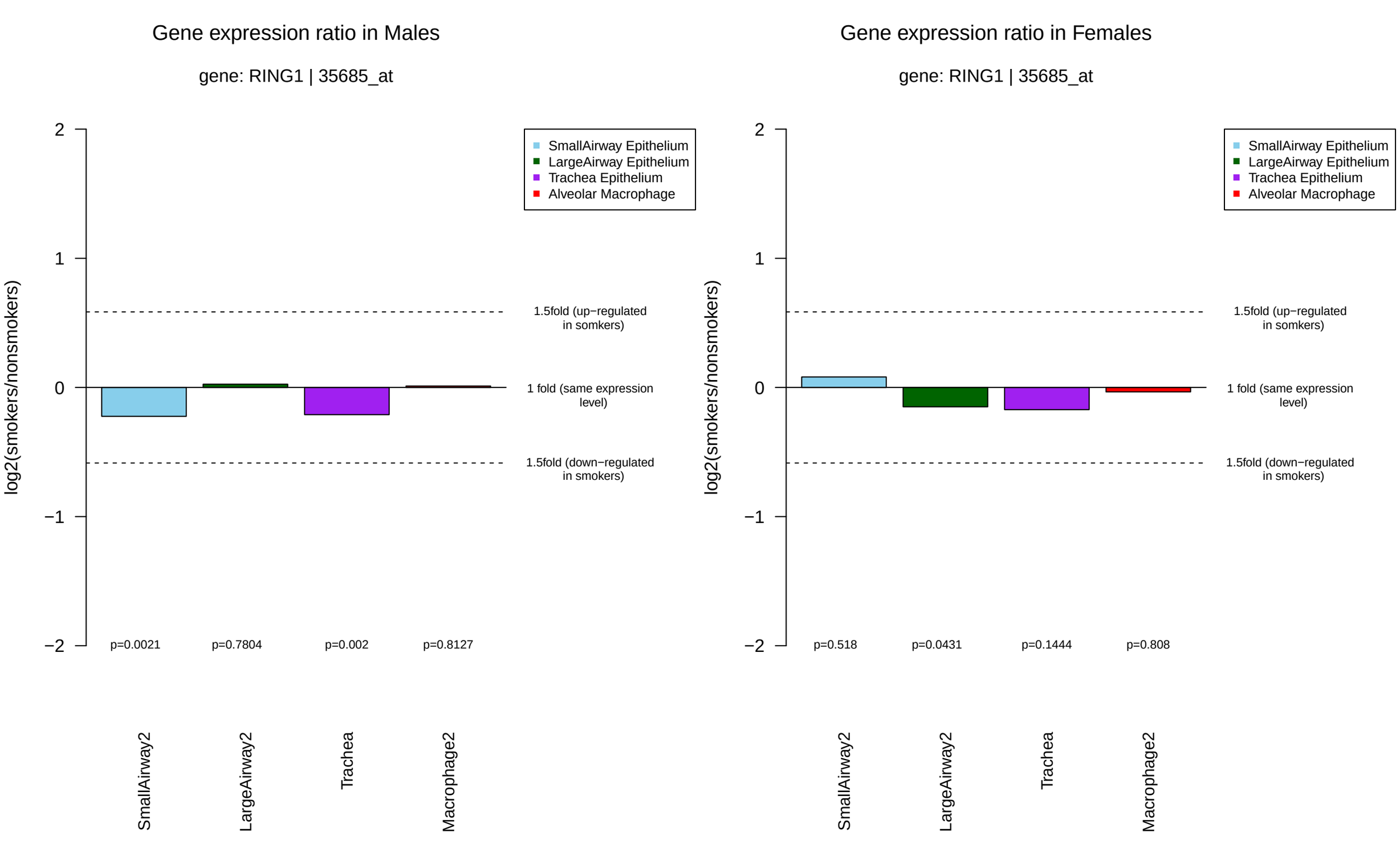
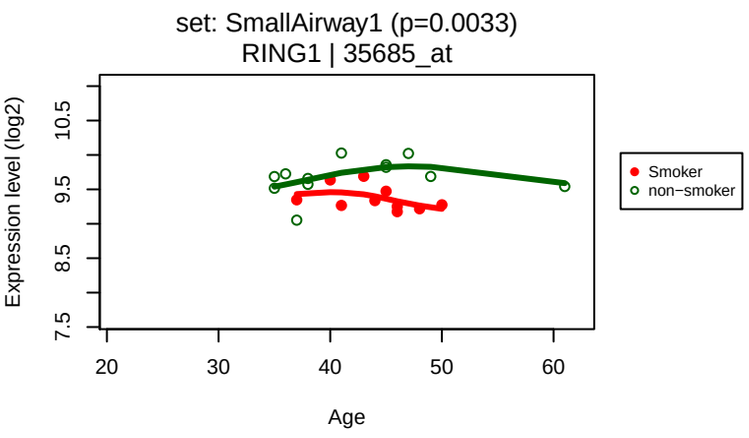
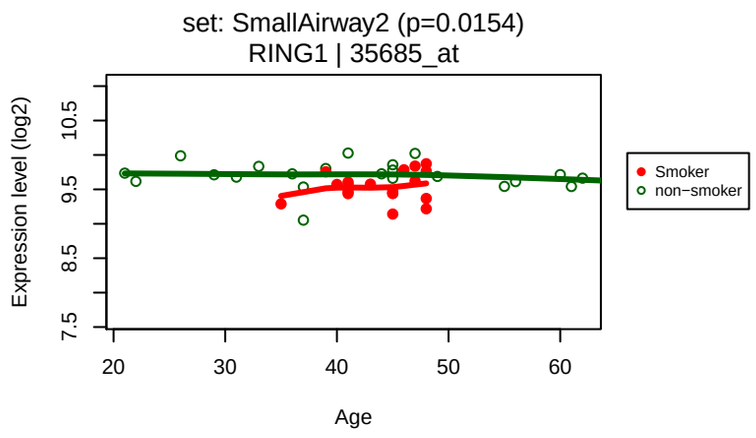
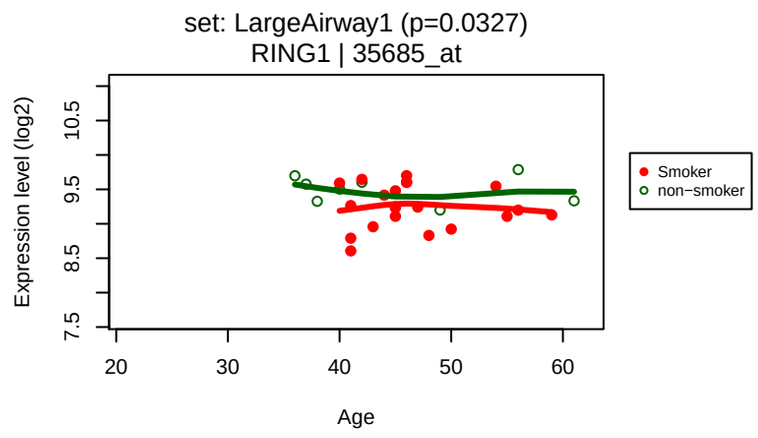

Gene expression of smokers and nonsmokers
gene: RING1 | 35685_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0033
p=0.0154
p=2e−04
p=0.0327
p=0.533
p=3e−04
p=0.2738
p=0.9949
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

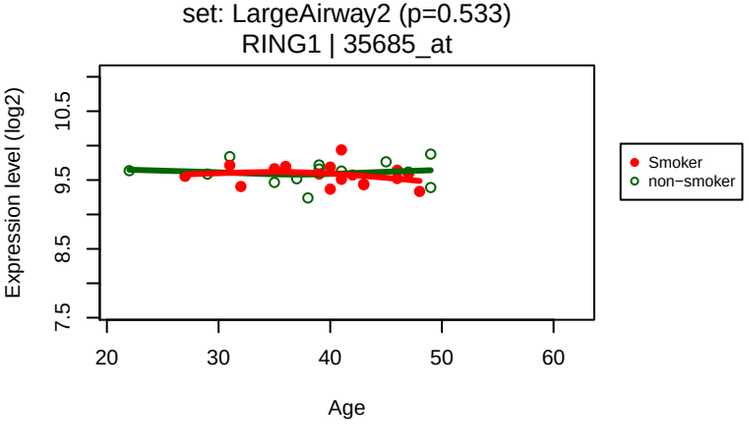

Gene expression ratio (smokers/nonsmokers)
gene: RING1 | 35685_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0033
p=0.0154
p=2e−04
p=0.0327
p=0.533
p=3e−04
p=0.2738
p=0.9949
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

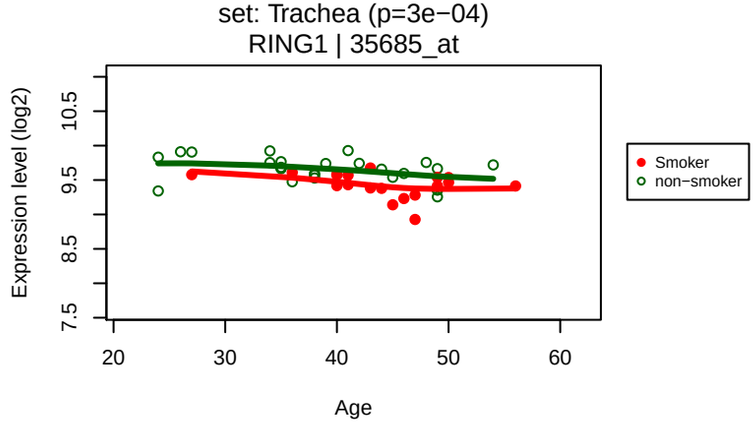

Gene expression ratio in Males
gene: RING1 | 35685_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.7804
p=0.002
p=0.8127
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RING1 | 35685_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.518
p=0.0431
p=0.1444
p=0.808
SmallAirway2
LargeAirway2
Trachea
Macrophage2

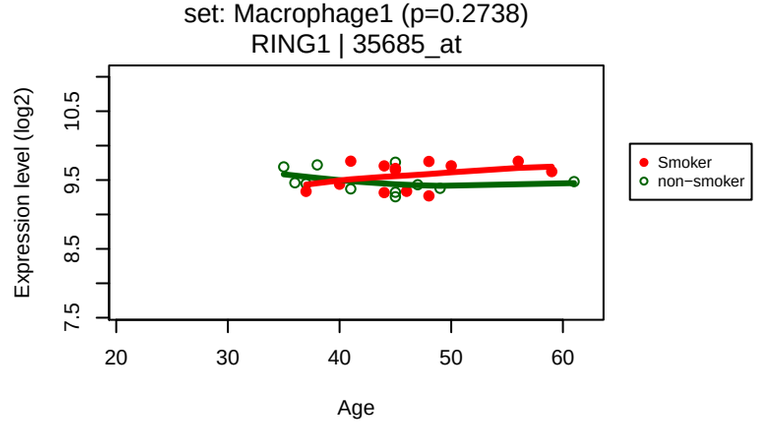

set: SmallAirway1 (p=0.0033)
RING1 | 35685_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

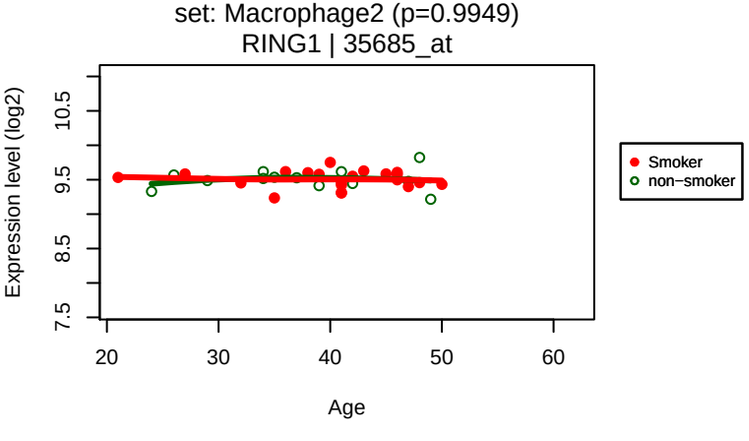

set: SmallAirway2 (p=0.0154)
RING1 | 35685_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

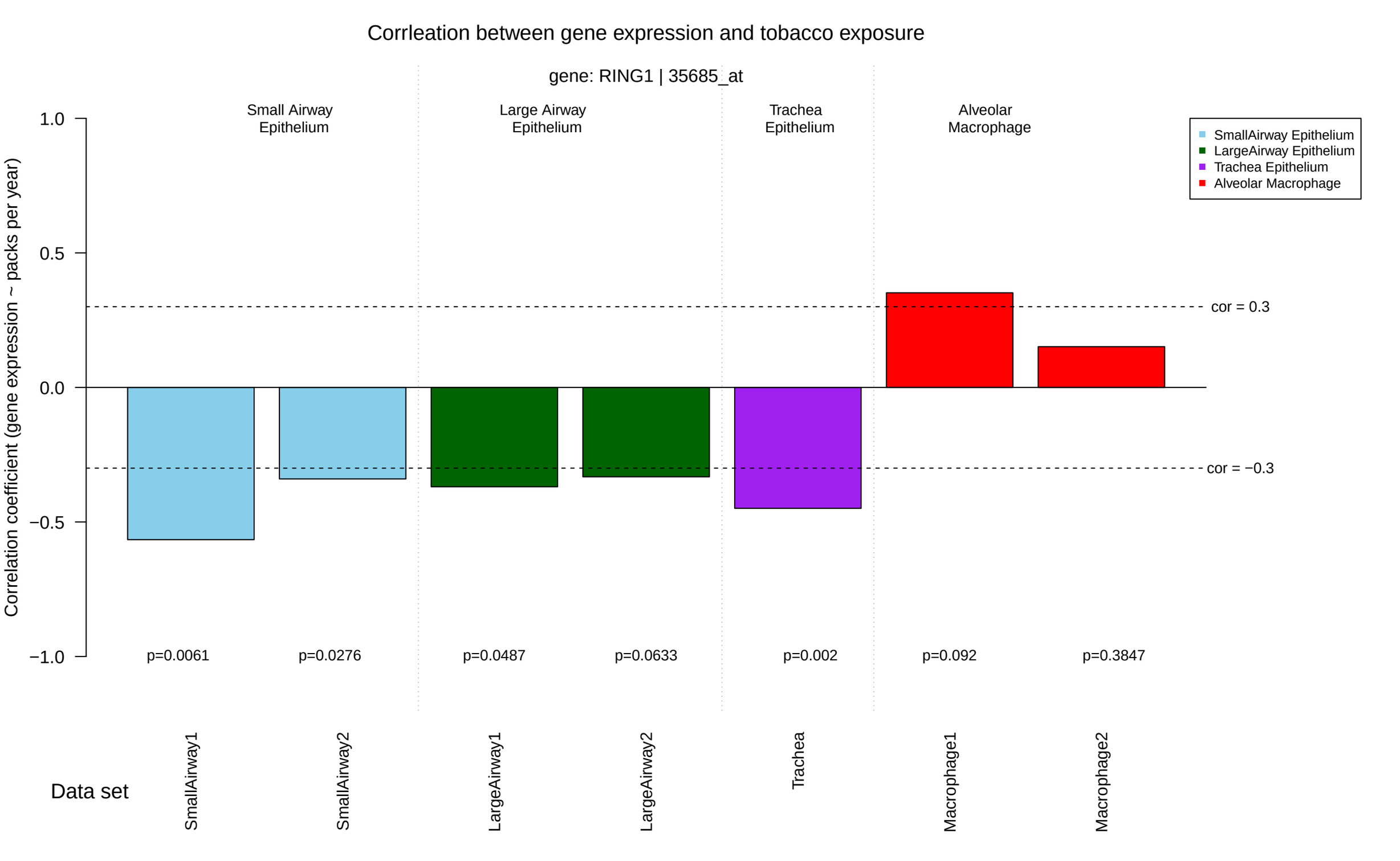

set: LargeAirway1 (p=0.0327)
RING1 | 35685_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.533)
RING1 | 35685_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
RING1 | 35685_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2738)
RING1 | 35685_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9949)
RING1 | 35685_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RING1 | 35685_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0061
p=0.0276
p=0.0487
p=0.0633
p=0.002
p=0.092
p=0.3847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2